Comparing Supercapacitors to Lithium-Ion Batteries Through Measurements and Simulations

Alain Abou Nakad [1], Mervat Madi [2], Olav Aaker [3], Efstratios L. Ntantis [4*], Karim Y. Kabalan [5]

[1] Mechanical Engineering Dept., Lebanese University, Hadath, Lebanon
[2] Electrical Engineering Dept., Higher Colleges of Technology Abu Dhabi, UAE
[3] Electrical Engineering Dept., Østfold University College, Halden, Norway
[4] Aerospace Engineering Dept., Amity University Dubai, Dubai, UAE
[5] Electrical Engineering Dept, American University of Beirut, Beirut, Lebanon
*entantis@amityuniversity.ae

Abstract: A vehicle powered by one or more electric motors is called an electric vehicle (EV). A battery, a collector system, or electricity from extravehicular sources can all be used to power it independently. Tesla cars are one of the most advanced electric vehicles. This study focuses on the comparison between Lithium-ion battery and supercapacitor, their characteristics, and their operation. The comparison was established using measurements and simulations in COMSOL Multi-physics software to investigate the most suitable for electric vehicles. The capacity fades of Lithium-ion batteries have been simulated and validated by actual measurements using a battery capacity tester. Finally, a new battery model is recommended as a solution for the long-time charging of batteries. This model is the Tanktwo string cells battery, distinguished by its fast charging time, flexible capacity, and low cost.

Keywords: Battery, Renewable Energies, Supercapacitors, Lithium-ion, Electric Vehicles, Tesla, Simulation and Experiment, COMSOL.

1. Introduction

Batteries are crucial in the global economic transition of maintaining a power system with a balance between supply and demand. Electrifying renewable vehicles (e-mobility), buildings, and cities using global conventional energy systems is the key to combating climate change and decarbonizing the world's transportation sector. Increased usage of solar, wind, hydropower, and other low-carbon technologies is a way to transition to a greener economy. These technologies include electric vehicles, storage capacity, microgrids, and smart grids. Eliminating greenhouse gas emissions from the environment attributable to electrification [1]. Today, clean energy technologies focus fundamentally on energy storage capacity. The charging rate of batteries is another crucial parameter of the power supply in the future of energy sustainability. The significance and applications of several battery types are covered in this article. Battery systems may update mini-grids, enable the "self-consumption" of rooftop solar power, and enable the energy storage in electric cars in addition to providing frequency response, reserve capacity, black-start capability, and other grid services [1].

A vehicle powered by one or more electric motors is known as an electric vehicle (EV). It can be independently powered by a battery, a collector system, or energy from extravehicular sources [2].

Electric vehicles (EVs) include, among other things, electric automobiles, rail and surface transportation, and aircraft. In the late 19th century, EVs appeared when electricity was a preferred motor vehicle propulsion type. At the time, gasoline-powered automobiles could not match the comfort levels and simplicity of operation that electric vehicles offered. Internal combustion (IC) engines prevailed as the main form of propulsion for vehicles and trucks for more than a century. In contrast, other forms of transportation, including trains and cars, continued to depend heavily on electric power [2].

This study refers to Tesla company efforts to improve EV batteries since from electric vehicles to battery technology, solar energy, energy storage, and more, Tesla has developed products that have helped to shift the world towards a more sustainable future and despite notable upticks in sales compared with a year ago, industry leader Tesla remains the top EV seller and has grown its lead over legacy automakers. With its corporate headquarters in Austin, Texas, Tesla is an electric vehicle and sustainable energy company. Tesla designs and manufactures electric cars, home and grid-scale battery energy storage, solar panels, solar roof tiles, and other goods and services. With 16% of the plug-in market and 23% of the battery-electric (pure electric) market share, the company sold the most plug-in and battery-electric cars. The goal of Tesla is to hasten the global switch to sustainable energy. The company's name honors Nikola Tesla, an inventor and electrical engineer. Tesla now creates solutions for renewable energy generation, storage, and fully electric automobiles. Since 2003, Tesla's investment in research for clean energy has been aiding the world's transition away from the use of fossil fuels and toward a future with no emissions. Its goal is to help accelerate the transition to sustainable transportation and energy provided by electric vehicles and solar panels. The Roadster debuted in 2008 and showcased Tesla's cutting-edge battery technology and electric powertrain. Then, Tesla built the world's first premium all-electric car from the ground up, the Model S, which soon won every automobile award in its category [3].

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This study examines the development of the devices that can be used in electric car, starting with Lithium-ion batteries. Following that, supercapacitors created by Maxwell Technology shall also be studied. The comparison between Lithium-ion batteries and supercapacitors was conducted with measurements and simulations to investigate the advantages and disadvantages of their performance and operation. The capacity fade of Lithium-ion batteries has been studied in this paper using the COMSOL multi-physics software and validated by experimental measurements with a battery capacity tester. The ideal use for future electric vehicles could be a new battery design called the Tanktwo string cells battery. Comparing or optimising the efficiency of different devices is frequently used through modelling. Hence, research on validating the modelling in COMSOL, ANSYS or MATLAB is continuing in [4-7] as well.

2. Device Models for Electric Vehicles

2.1. Supercapacitors

Supercapacitors (SCs) are electrochemical devices that can store and provide high-power electricity fast for an extended period of time. The most basic supercapacitor is made up of two electrodes with an electrolyte placed in between. No chemical oxidation-reduction processes exist; the electrical charges are organized at the electrode/electrolyte contact. Materials must have a large surface area to collect many electric charges since the physical mechanism of accumulation is restricted [8].

Table 1 Characteristics of Supercapacitors

Function	Supercapacitor
Charge time	1-10 seconds
Cycle life	1 million or 30,000h
Cell voltage	2.3 to 2.75V
Specific energy (Wh/kg)	5
Specific power (W/ kg)	Up to 10,000
Cost per Wh	20$
Service life	10 to 15 years
Charge temperature	-40^{0}C to 65^{0}C
Production company	Maxwell Technologies

Table I presents the characteristics of supercapacitors. A double-layer capacitor with a very high capacity and low voltage restrictions is known as a supercapacitor. Supercapacitors feature larger plates than electrolytic capacitors and can hold more energy. They are appropriate for applications operating in the 1.8 V to 2.5 V voltage range and have a low leakage current. A supercapacitor has a lifespan of 10–20 years, although, after around 8–10 years, its capacity might drop from 100% to 80% [8].

Two electrodes are separated in supercapacitors by an ion-permeable membrane (separator). An electrolyte fills the space and ionically connects the electrodes. They do not rely on chemical reactions. Instead, they store potential energy electrostatically within them. Supercapacitors use dielectric or insulator between their plates to separate the collection of positive (+ve) and negative (-ve) charges building on each side's plates. It is this separation that allows the device to store energy and quickly release it. It basically captures static electricity for future use. The most significant advantage of this is that a 3V capacitor now will still be a 3V capacitor in 15-20 years. In contrast, on the other hand, a battery may lose voltage capacity over time and repeated usage. In the electrolyte, ions create electric double layers with a polarity different from the electrodes when an applied voltage polarizes the electrodes. For example, a layer of negative ions will be present at the electrode/electrolyte interface of the positively polarized electrode. A layer of positive ions will adsorb onto the negative layer to balance the charge. The negatively polarized electrode exhibits the opposite behavior [9].

The self-discharge rate is the fundamental drawback of supercapacitors. Supercapacitors are inadequate for storing energy for a long time. Compared to Lithium-ion batteries, supercapacitors drop considerably; they can lose up to 10 to 20 percent of their charge daily due to self-discharge [9]. Supercapacitors do not use conventional solid dielectric. Supercapacitors can meet in three types: electrostatic double-layer capacitance, electrochemical pseudo capacitance, and hybrid capacitors, which contribute to the total capacitance of the capacitor with few differences [10]. There are several ways to make supercapacitors, including flat devices with single electrodes looped around a cylindrical case or stacks of devices housed in a rectangular case [8].

2.2. Lithium-ion Battery

A particular type of rechargeable battery is a Lithium-ion (Li-ion) battery used in numerous portable electronics and electric vehicles. Because they possess a higher energy density compared to standard lead-acid or nickel-cadmium rechargeable batteries, battery makers may conserve space by making battery packs that are smaller overall. The requirement for next-generation Li-ion batteries with superior safety performance, high power, high capacity, rapid charging rate, long life, and low cost per watt-hour energy is another significantly increasing market for electric and hybrid cars. Lithium-ion (Li-ion) batteries power the Tesla Roadster model car. The success of the Tesla company is mainly due to Lithium-ion batteries having the highest charge capacity of any battery composition ever developed at a low cost enough to make battery electric vehicles practical [11].

A Lithium-ion battery comprises an anode, a cathode, a separator between the two electrodes, and an electrolyte that takes up the remaining battery area. The anode and cathode may hold Lithium ions, which release and store energy as they flow between the electrodes through the electrolyte. Positively charged Li+ ions pass from the cathode through the electrolyte and over a separator into the anode during the Li-ion battery charging cycle to balance the flow of electrons in the external circuit. When the battery discharges and generates an electric current, the anode transfers Lithium ions to the cathode, causing an electron flow from one side to the other [12].

Table 2 Characteristics of Lithium-ion battery

Function	Lithium-ion
Charge time	10-60 minutes

Cycle life	500 and higher
Cell voltage	3.6 to 3.7V
Specific energy(Wh / kg)	100-200
Specific power (W / kg)	1000-3000
Cost per Wh	0.5-1 $
Service life	5 to 10 years
Charge temperature	0 to 45^{0}C

The maintenance, longevity, charging speed, safety, and convenience are the five primary characteristics under which Lithium-ion battery benefits may be divided, as shown in Table II. They outperform all other rechargeable batteries in terms of stability, energy density, voltage capacity, and self-discharge rate, and they may be recharged hundreds of times [13].

3. Comparison Between SCS And Li-Ion Batteries

3.1. Analytical Comparison

Comparing the Lithium-ion battery with a supercapacitor, the Lithium-ion battery can store more energy per unit mass because it has higher energy density. Still, the supercapacitor has a higher power density to keep and release the power in a shorter term. The high energy density makes supercapacitors the best choice for storing and releasing high power faster. However, Lithium-ion batteries are still the most appropriate for storing significant amounts of energy over long periods [14]. Compared to Lithium-ion batteries, which have a limited cycle life, supercapacitors potentially have an unlimited capacity for charging and discharging. A supercapacitor's self-discharge is significantly more than a Lithium-ion battery, which causes a linear discharge curve. The linear discharge curve is considered a critical disadvantage for supercapacitors. Supercapacitors are costly in terms of cost per watt and have low specific energy. Some design experts contend that a larger battery would be a better use of the supercapacitor's price [15]. Lithium-ion batteries are the best batteries for electric cars since ultra-capacitors cannot simultaneously meet the systems for EVs' high-power and high-energy supplies. Consequently, Lithium-ion batteries provide long-term self-reliance for the vehicle while the ultra-capacitors manage short-term power needs. For instance, while the traction batteries used by Tesla are all Lithium-ion batteries, they differ. Each of the major cathode chemistries, which are many, has changed throughout time [16].

Table 3 Comparison between supercapacitors and Lithium-Ion batteries

CHARACTERISTIC	LI-ION	SUPERCAPACITOR
Stored energy	Higher	Lower
Instantaneous power	Lower	Higher
Charge-recharge cycles	Lower	Higher
Charging speed	Slower	Faster
Lifetime	Shorter	Longer
Cost per watt	Higher	Lower
Cost per watt-hour energy	Lower	Higher
Weight	Lower	Higher
Operating temperature	About -20 to 60 °C	About -40 to 85°C
Discharge curve	Long constant voltage	Linear discharge

3.2. Simulation Results and Comparison

In this part, using COMSOL Multi-physics software, the two battery types, Lithium-ion and supercapacitors, that have already been discussed before are analyzed.

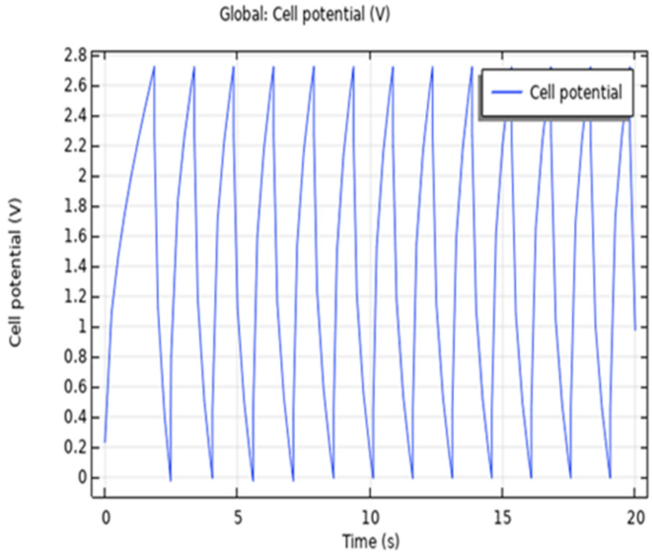

Fig. 1. *Charge and discharge cell potential of Supercapacitors in the function of time*

Figure 1 analyses the cell potential of a supercapacitor during numerous charge-discharge cycles in the function of time. The cell potential increases to reach its maximum at 2.7 V and then decreases during discharge to get the minimum at 0 V. In twenty seconds, twelve charge-discharge similar cycles occurred.

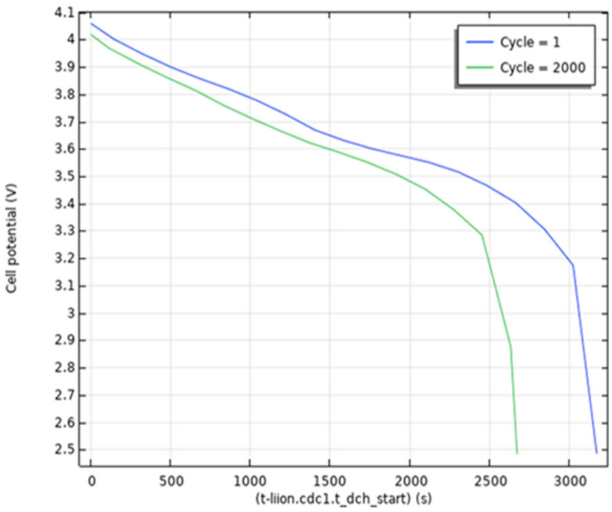

Fig. 2. *Cell voltage of Lithium-Ion during discharge in the function of time for cycles 1 and 2000*

Figure 2 shows the Lithium-ion battery's cell voltage as a function of time for cycles 1 and 2000. In cycle 1, the

Lithium-ion battery voltage decreases from 4.1 V to 2.5 V within 3600 seconds. While in cycle 2000, the battery voltage drops within 2600 seconds.

Comparing Figures 1 and 2, we conclude that the charge-discharge remains the same for many cycles in a supercapacitor. On the other hand, the Lithium-ion battery's charge-discharge cycle needs a longer time than in a supercapacitor. As a result, a Lithium-ion battery's charging and discharging life cycle is much longer than a supercapacitor.

Another difference that remains is the discharge curve of the two devices. Supercapacitors, compared to Lithium-ion batteries, have a linear discharge curve. One of the primary disadvantages of supercapacitors is the linear discharge curve obtained due to high self-discharge.

This disadvantage is one of the reasons which let electric vehicle companies use only Lithium-ion as a battery pack, such as in the case of Tesla cars. For this reason, the actual measurement is studied concerning the capacity fade of Lithium-ion batteries to validate COMSOL results.

4. Capacity Fade of Li-Ion Battery

4.1. COMSOL Software

Regarding battery performance, both the loss of cyclable Lithium and the loss of active materials cause the battery's capacity to decrease. The cyclable Lithium is consumed to form passivating films on electrode materials during the Lithium-ion cell's formation cycles, consisting of one to several charge-discharge processes. Because of the cyclable Lithium loss, each Lithium-ion cell loses 10 to 15% of its capacity [17]. As shown in Figure 3, the relative capacity of Lithium-ion batteries based on cyclable Lithium decreases with time from 1 to 0.87. On the other hand, after a 1C discharge capacity, i.e., 1 hour of discharging, the relative capacity of the battery decreases from 1 to 0.88. Then it continues to fall more and more with time. The following formula defines a Lithium-ion battery's relative capacity:

$$Q_R = (Q_0 - Q_{SEI})/Q_0 \qquad (1)$$

Where Q_R and Q_0 are the relative and initial capacity of the battery, respectively, and Q_{SEI} is the capacity after discharging.

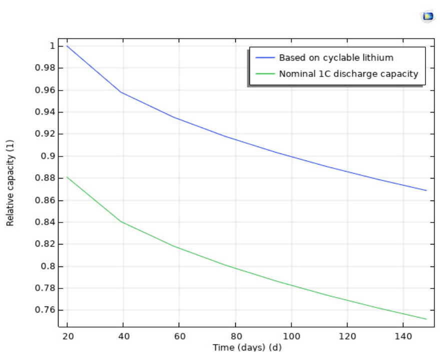

Fig. 3. *Capacity of Li-ion in function of time during discharge*

4.2. Validations and Measurements

4.2.1 Description: The device used is a 18650 Li-ion Battery Capacity Tester and Resistance Lead-acid Battery Capacity Meter Discharge Tester showed in Figure 4. Single-function, by controlling the load off and getting real-time integrated discharge current capacity results. Maximum support 15V 3A discharged through the discharge voltage and the current itself does not discharge. Maximum capacity statistics support 9999Ah (9999000mAh). Automatically identify and develop appropriate battery termination voltage can also be manually adjusted.

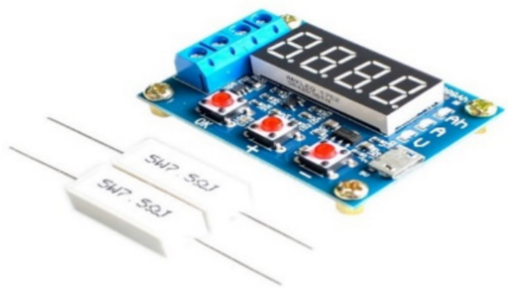

Fig. 4. *Zb2l3 battery capacity tester*

Figure 5 shows the installation of the tester and the battery. The first step is to connect the battery under test: the positive terminal of the battery to the positive input (red wire) of the tester and the negative terminal to the negative input (black wire). The load is connected to the positive and negative outputs of the tester's micro USB power source.

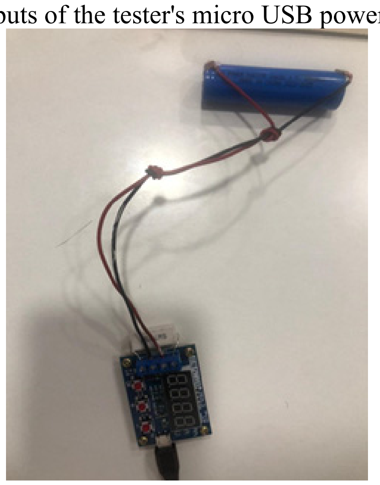

Fig. 5. *Installation of the tester and battery*

The tester can automatically produce an appropriate termination voltage of the battery's full charge voltage. It will begin flashing thrice following the test with a single "OK" button touch.

The load control is disabled whenever the battery voltage reaches the chosen cut-off value and the appropriate indicator flashes rapidly. It now displays the battery's true capacity as discharging capacity. Pressing the "OK" button to stop the flashing allows for a steady data display; pressing the "OK" button again to return to the power-on state can substitute the battery test in the next section.

4.2.2 Results: In this study, the zb2l3 battery capacity tester is used to discharge a Lithium-ion battery of 4000 mAh and 3.7 V. At first, the battery's voltage is 3.7 V. We click on the "–" button to indicate the dropped voltage we need, i.e., 3 V. The target is to see the battery's capacity after discharge from 3.7 V to 3 V. After 50 minutes of discharging from 3.7 V to 3 V, the capacity reduces from 4Ah to 0.68Ah. The relative capacity given in Eq.1 is applied and calculated as follows:

$$Q_R = (Q_0-Q_{SEI})/Q_0 = (4-0.68)/4 = 0.83 = 83\%$$

So the discharging of a Lithium-ion battery from 3.7 V to 3 V takes 50 minutes as time and 83% as relative capacity.

4.3. Results Comparison

Table IV shows the characteristics of a Lithium-ion battery as studied using COMSOL Multi-physics software and validated with experimentation. The obtained results are similar, which validates the current study, and the slight error is related to the difference between the voltage taken into consideration in both cases.

Table 4 Results comparison between COMSOL software and measurements

Characteristics	COMSOL Multi-physics	Validation
Voltage (V)	4 – 2.5	3.7 - 3
Time (minutes)	60	50
Relative capacity (%)	0.88	0.83

5. Future Scope

Tanktwo investigated the possibility of modularizing a battery to convert hours into minutes. A vehicle's battery pack is replaced with a container containing thousands of tiny, intelligent string cells in Tanktwo's technology. Each string cell comprises a conductive chemical-coated plastic container, enabling it to contact neighboring cells quickly and effortlessly. The battery's small balls are sucked out and switched for recharged cells at a service station to allow faster charging in an electric vehicle. The batteries can be charged at dedicated stations. In the tank, the String Cells form electric circuits and act as a single battery pack. When they run out of charge, they can be swapped out in just minutes to eliminate lengthy (and costly) charge time that keeps commercial vehicles idle [19]. The Tanktwo system consists of four main components: String Cells, String Tank, Cells Swap Station, and Tanktwo Cloud. The string battery consists of several electrochemical string cells. The string battery determines the optimum route to link the string cells after mapping the string cells. A string cell's electrochemical cell has positive and negative polarity. The internal processing unit of the string cell connects any of the contacts on the surface of the string cell to either the positive or negative terminal of the internal cell. The inner cell generates the electric power, but using the routing instructions received from the string battery, the string cell sends electricity to any external connections [20]. The string cell technology reduces the average EV battery cost by forty to eighty percent compared to today's best-in-class technology by utilizing the possibility of defining the pack individually [21]. In less than three minutes, vehicles exchange a tank of uncharged cells for charged cells. Users of the Tanktwo battery may customize their battery packs to suit their driving requirements. Reducing the cost, weight, and daily energy use through battery optimization [22]. The Twotank string cells battery represents the solution to electricity storage for EVs and similar applications in the near future. The Tanktwo String battery overcomes the current issues with EV batteries, including their high initial costs, lack of use, and sluggish charging times. It also makes building new EV charging stations more manageable and cost-effective [23].

6. Conclusion

Electricity is the way of the future [1]. Certainly, the universe will soon be powered mostly or entirely by electricity. As a result of the increasing demand for batteries, there should be strategies to reduce prices to increase global production of batteries and electric technologies. The fastest-growing storage technology, batteries, will be necessary to meet the European Union's goal of a 55% reduction in greenhouse gas emissions by 2030 [1]. Supercapacitors and Lithium-ion batteries were studied while focusing on the characteristics of each one. In this study, we demonstrated that Lithium-ion battery is more efficient and can be better for electric cars, especially for Tesla cars. Lithium-ion (Li-ion) batteries are available in various electrode materials, making this a rich, sophisticated, and ever-flexible technology. Performance improvements are always conceivable, with several parameters that may be improved. Different applications may be developed depending on their characteristics, preferentially requiring energy, power, longevity, etc. The greater benefit of a certain energy determines the availability of additional power. Because they have greater specific energy than supercapacitors, Lithium-ion batteries deliver more power. Supercapacitors have more specific power than Lithium-ion batteries because they have low resistance, which results in the loss of usable energy. Supercapacitors, on the other hand, are becoming increasingly popular in many energy industries, where continual high energy is mostly needed for computer chips, portable electronic devices, and toys like automobiles and drones. Supercapacitors cannot store the greatest amount of energy. Hence their usage in backup systems is limited [18]. When it comes to finding a way to charge EVs more quickly, human ingenuity has developed some pretty wide-ranging solutions, from battery-swapping to flow batteries that store energy in tanks of liquid electrolytes. The latest innovative idea comes from startup Tanktwo, and is called a String Battery.

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